Quantifying leaf angle and its relationship with water availability in two California oak species

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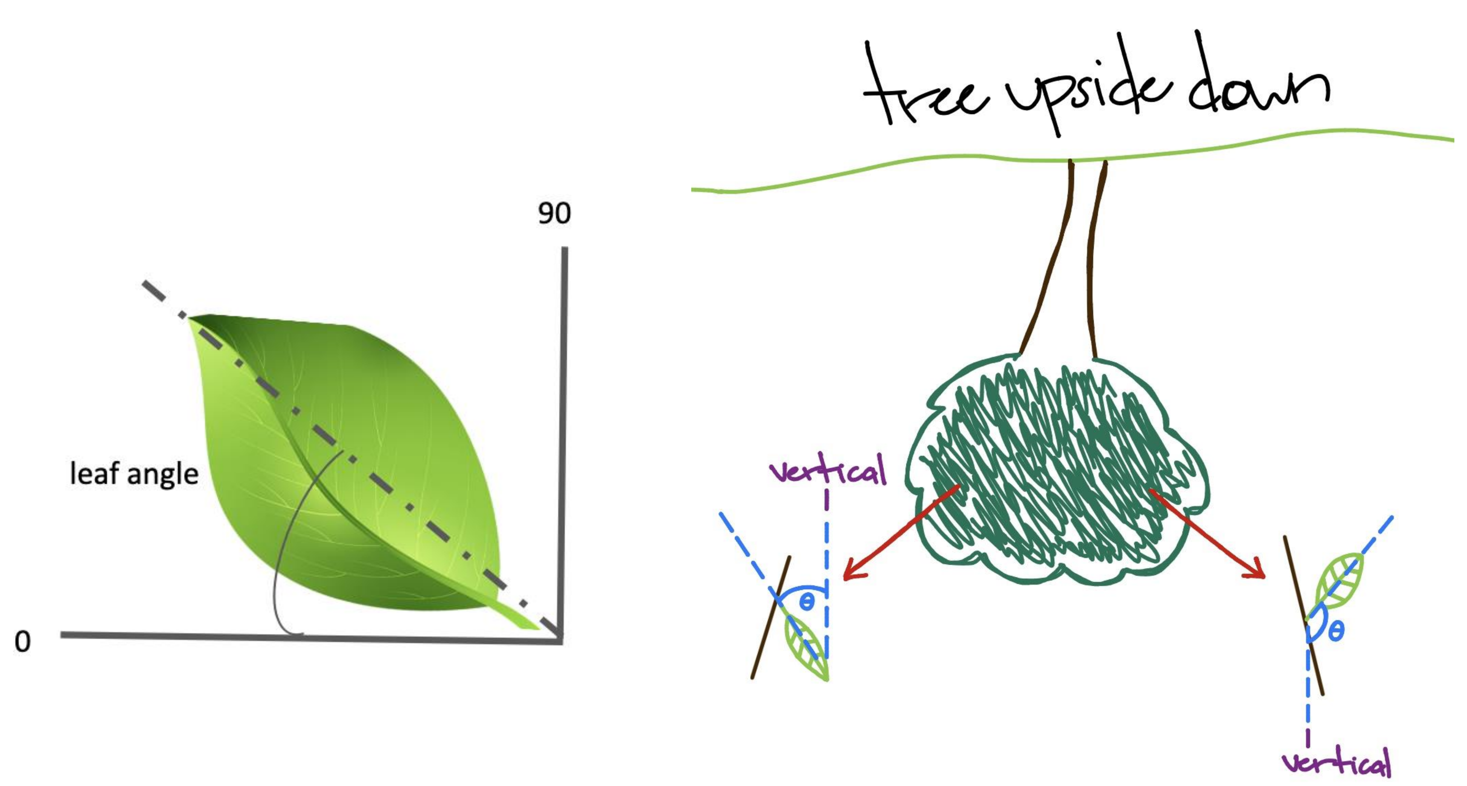
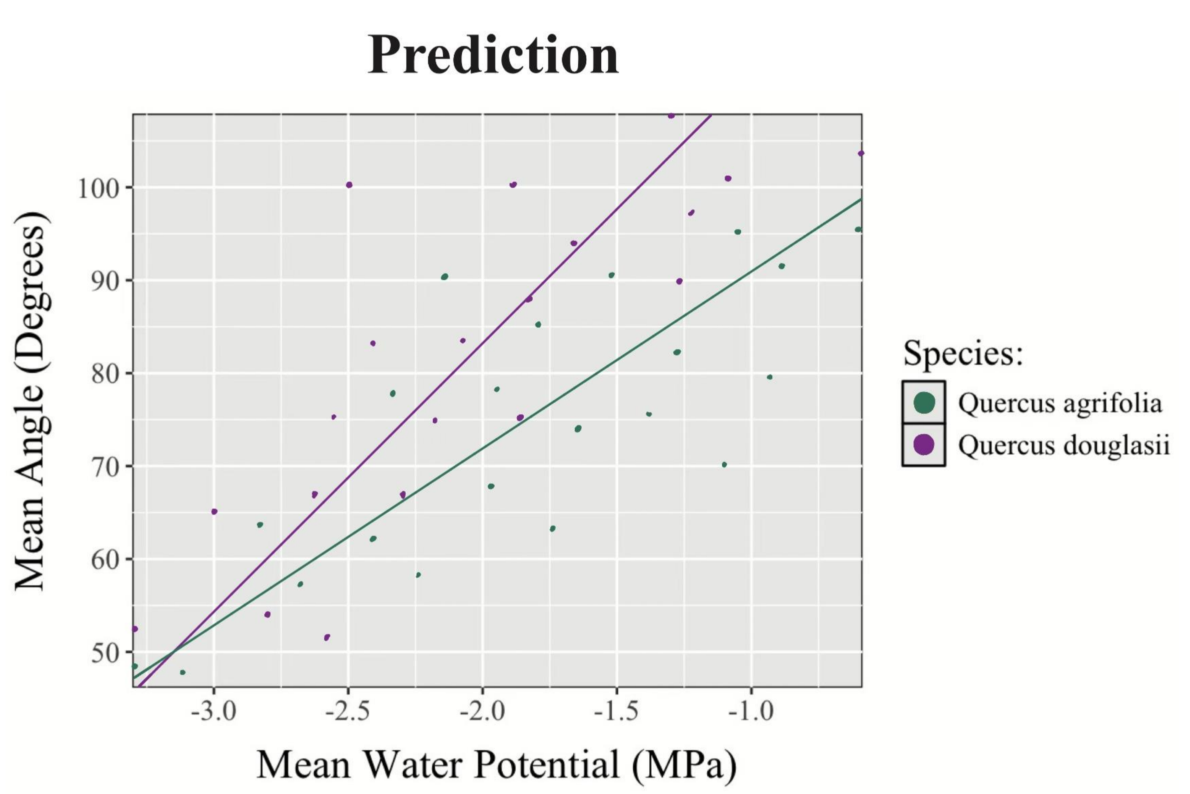
BACKGROUND:

- Climate change has led to rising temperatures and more severe droughts in California and worldwide
- Native California oak species are especially at risk
- Leaf angle may be adaptive in hot/dry climates: aligning leaves to avoid direct midday sun exposure plants can escape daily maximum temperatures
- Water shedding from leaves may be due to leaf angle
- Leaf angle adjustment in oak species may be an adaptation to high drought stress conditions

AIMS & QUESTIONS:



Aim: Develop a protocol for measuring leaf angle from oak species at Sedgwick reserve.



Question:

1. Do trees with lower mean water potential (increased drought stress) exhibit higher mean leaf angles (to avoid hot temperature exposure)?

2. Do two oak species with differing leaf habits (evergreen vs. deciduous) exhibit different leaf angle?

METHODS:

- Gathered photographs from 10 coast live oaks, *Quercus agrifolia*, and blue oaks, 10 *Quercus douglasii*
- Selected 2 images/tree and measured 10 leaves/image
- Recorded azimuth, pitch, and angle of photo taken given
- Used ImageJ to measure angle related to straight line (leaf angle) for each of the 10 leaf samples per image

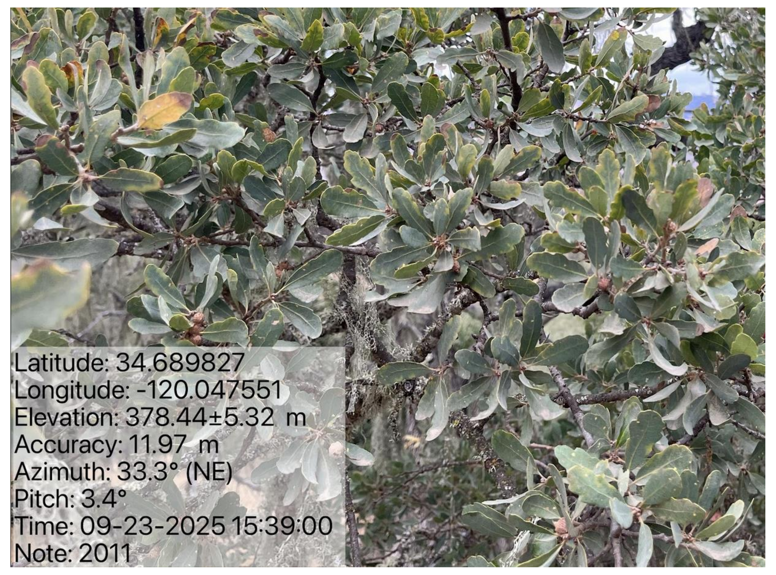


Figure 1. Level photos of the mid-canopy of sample *Q. douglassii* oak with coordinates, angles, date, and tree number.

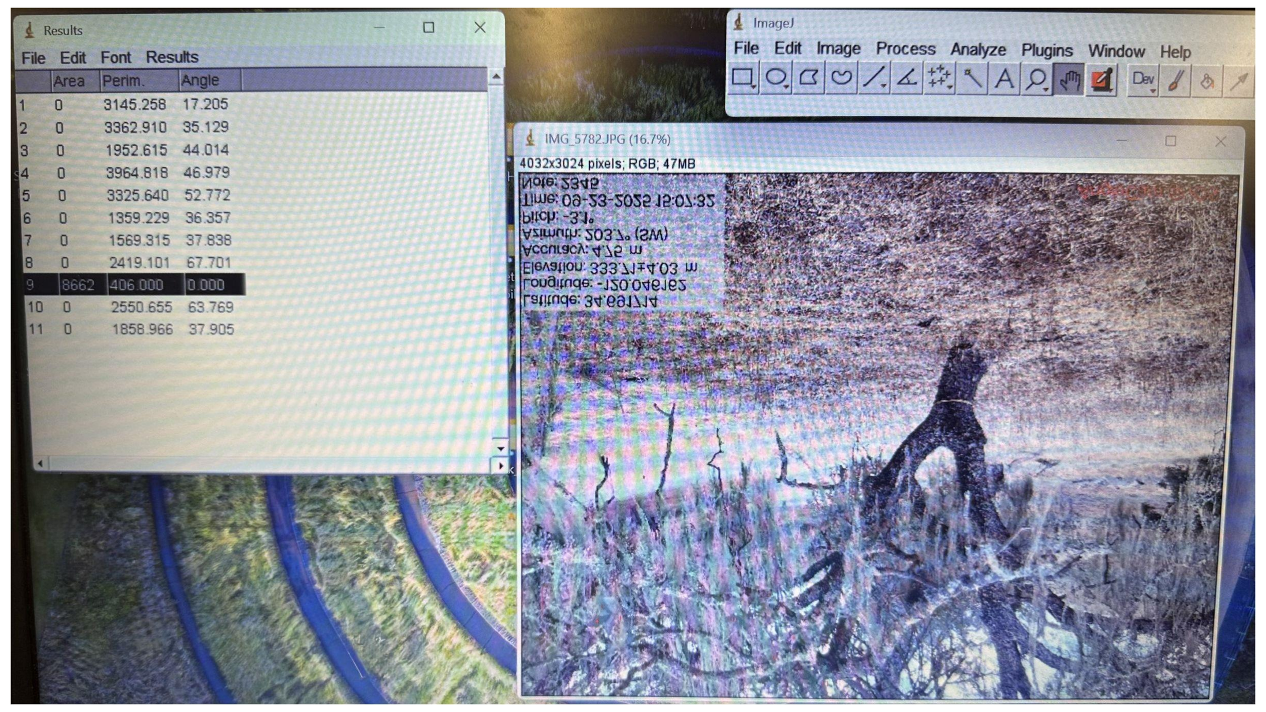


Figure 2. Oak image (flipped upside down) on ImageJ with ten leaf angle measurements taken (top left).

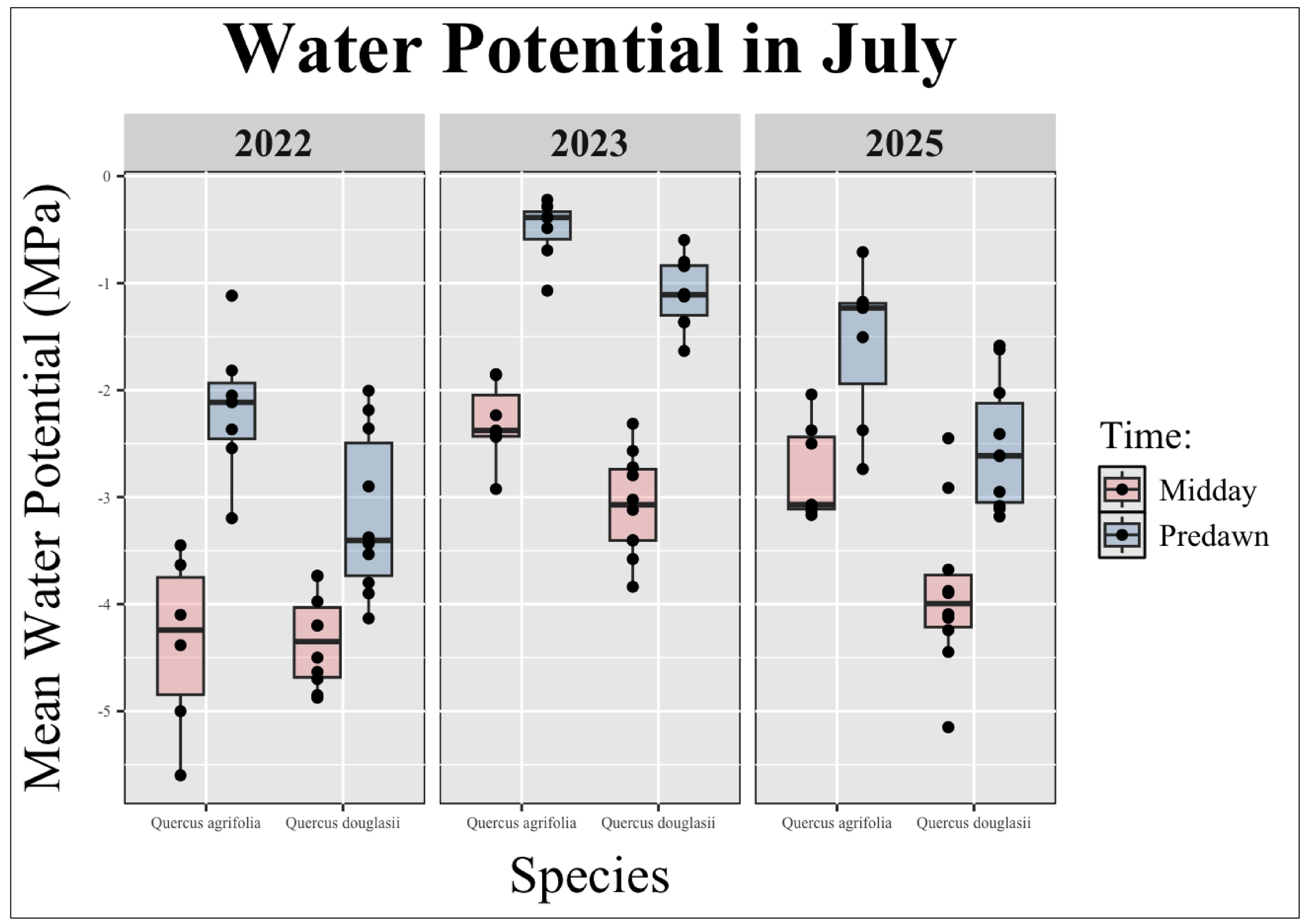


Figure 3. **Species exhibit different levels of drought stress across years in July.**

RESULTS:

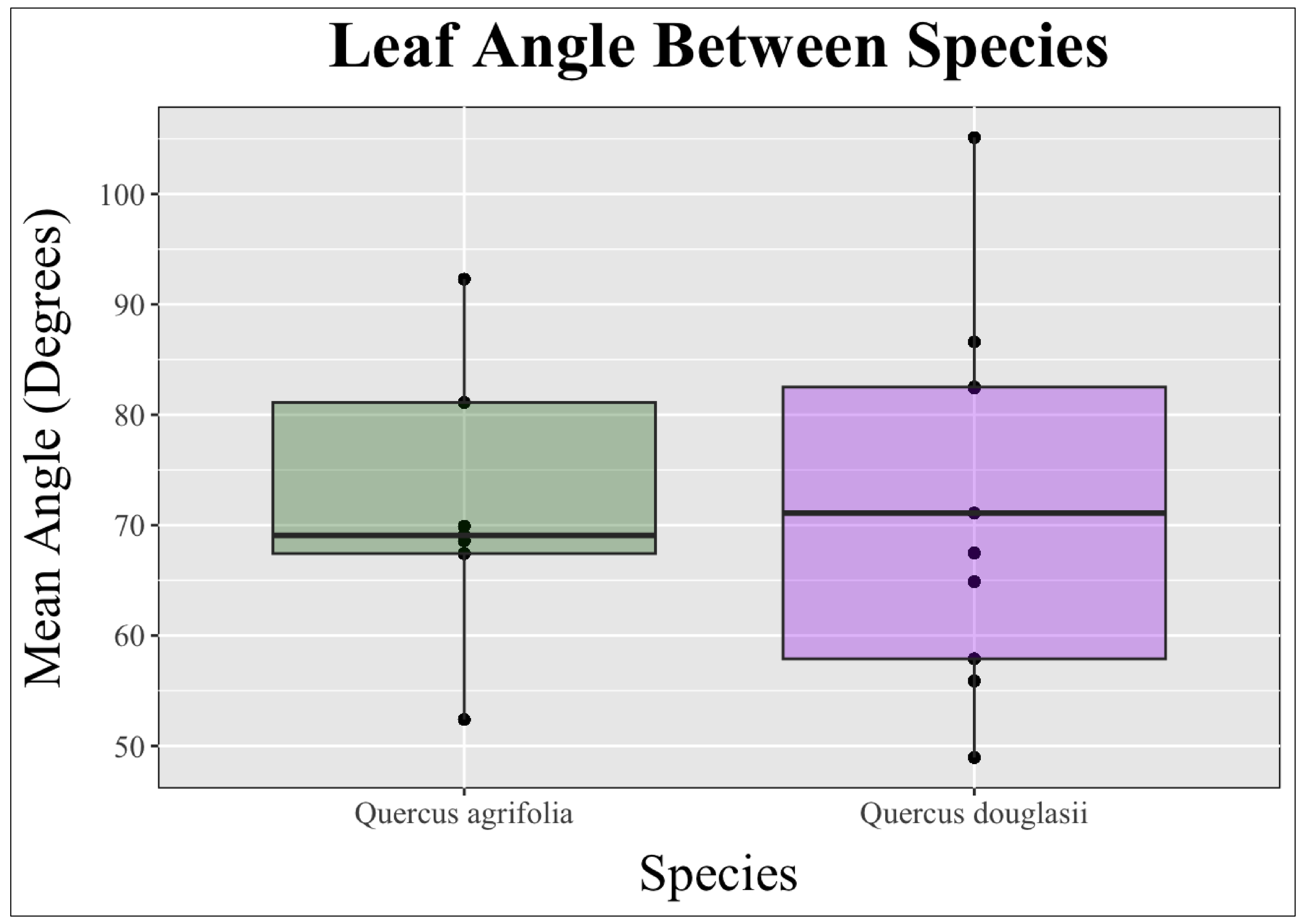


Figure 4. **Species show no significant difference in average leaf angle.**

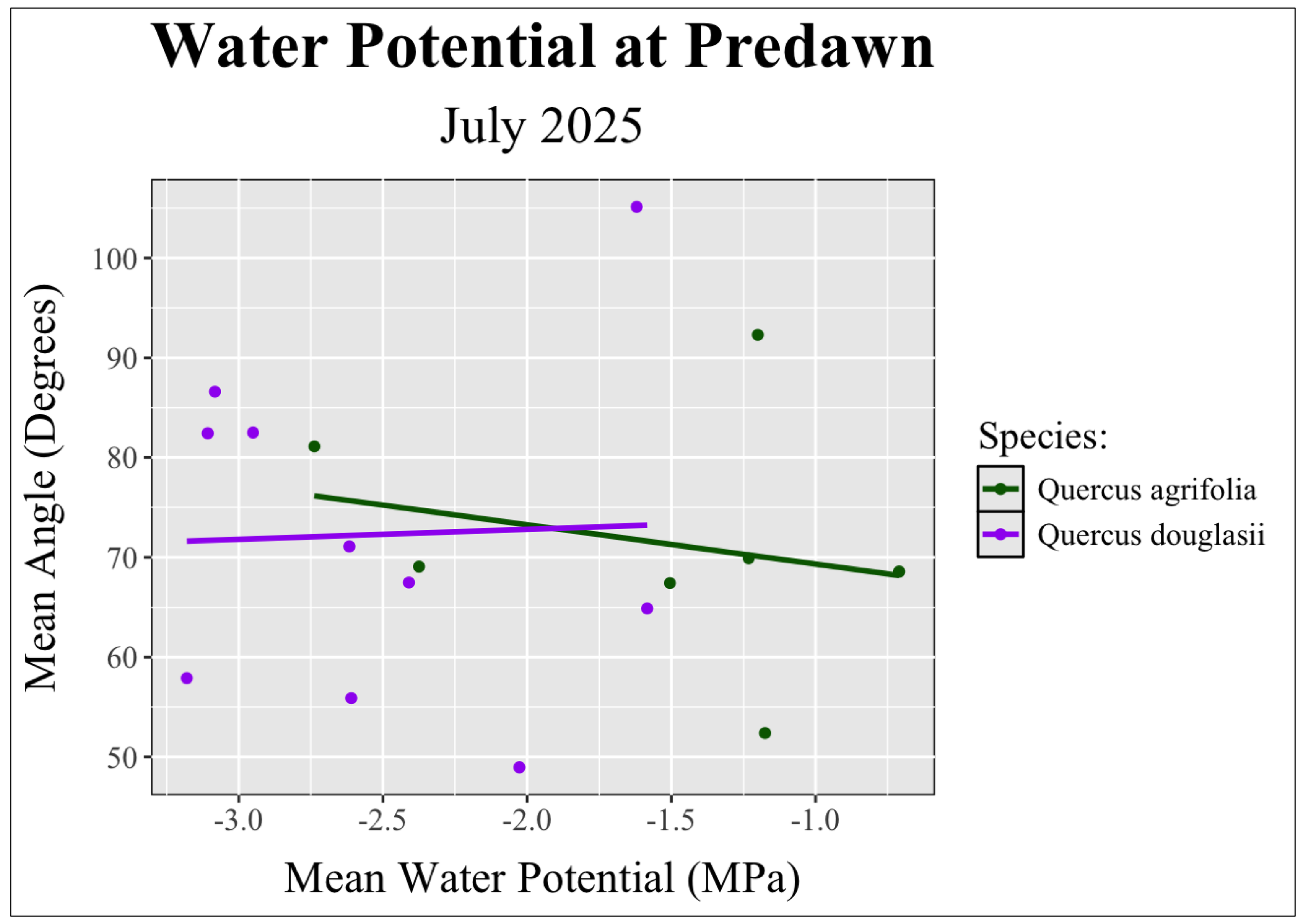


Figure 5. **Species show no significant difference between levels of drought stress in July 2025 and average leaf angle.**

CONCLUSION:

- No relationship between leaf angle and drought stress
- Blue oaks show greater drought stress than coast live oaks
- High variability in leaf angle for each individual tree
- Need more data collection and samples of leaf angle
- Photographic methods are effective for measuring individual leaf angles but are inefficient for larger samples

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2. Li, S., Fang, H., & Zhang, Y. (2023). Determination of the Leaf Inclination Angle (LIA) through Field and Remote Sensing Methods: Current Status and Future Prospects. *Remote Sensing*, *15*(4), 946. https://doi.org/10.3390/rs15040946
3. McNeil, B. E., Pisek, J., Harald Lepisk, & Flamenco, E. A. (2016). Measuring leaf angle distribution in broadleaf canopies using UAVs. *Agricultural and Forest Meteorology*, *218-219*, 204–208. https://doi.org/10.1016/j.agrformet.2015.12.058